# Simulated Data and Dataset Statement

**Project:** Real-Time Network Intrusion Detection Using Machine Learning-Based Multi-Class Classification
**Group:** Group 19 | FUSE AI-201

## Dataset used

The model training and benchmark evaluation use the publicly available NSL-KDD dataset, specifically KDDTrain+ and KDDTest+, obtained in the Module 5 notebook through KaggleHub (hassan06/nslkdd). The dataset consists of labelled network connection records with 41 input features. It is public benchmark data, not a newly collected live network traffic feed and not a dataset of current traffic from an operational organization.

## Synthetic data use

The repository includes one constructed illustrative CSV record to demonstrate the expected 41-feature input format for the prediction interface. This sample is not claimed to be an observed network event or an independently measured traffic record. During model training, SMOTENC generates synthetic minority-class examples within the training pipeline to address class imbalance. Those synthetic examples are used only for fitting the model; the validation split and official KDDTest+ evaluation records are not oversampled.

## Privacy and limitations

No private packet captures, customer traffic, credentials or personally identifying network records are included in this project package. The benchmark is not representative of every current enterprise network environment. Results on NSL-KDD must not be interpreted as evidence that this prototype detects modern attacks, zero-day attacks or live network incidents reliably.

## Source and attribution

The notebook downloads KaggleHub dataset hassan06/nslkdd. A public NSL-KDD mirror is documented at https://github.com/jmnwong/NSL-KDD-Dataset. Original reference: M. Tavallaee, E. Bagheri, W. Lu and A. A. Ghorbani, “A Detailed Analysis of the KDD CUP 99 Data Set,” IEEE Symposium on Computational Intelligence for Security and Defense Applications, 2009. The raw benchmark files are not redistributed in this repository; users must review and follow the dataset host’s usage and redistribution terms.

## Declaration

Accordingly, this project uses public benchmark data for training and evaluation, includes a constructed sample input for interface demonstration, and uses SMOTENC-generated synthetic observations for minority-class training only. No model-performance claim is based on a live operational dataset.